



Total Marks: 50

Time: **1 Hour 30 Minutes**

- There are **four(4)** questions. Question **1** is **mandatory**. Answer **any two** from the remaining.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet.

Name:	ID:	Section:
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Question 1: [Mandatory]

[20]

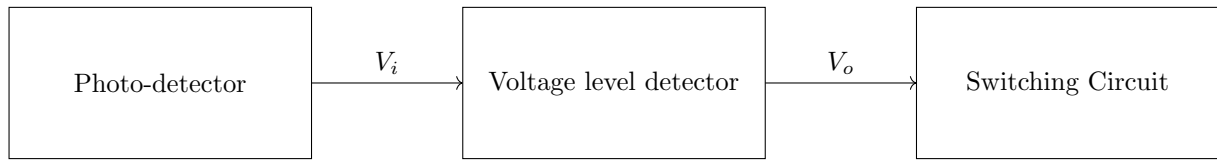
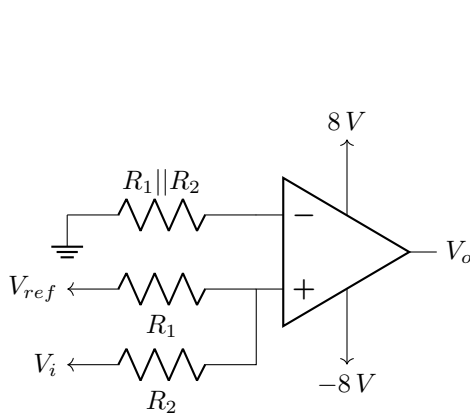
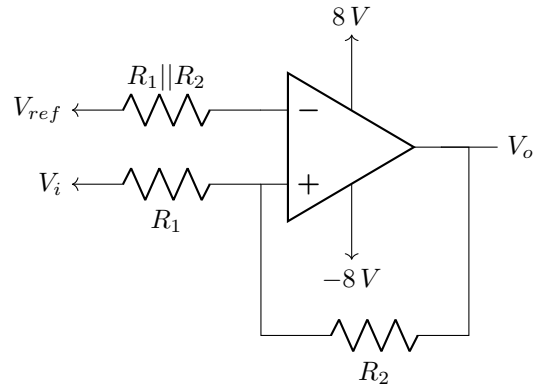


Figure 1: Flow diagram of a photo-detector switching system



(a) Comparator realization of Voltage level detector



(b) Schmitt Trigger realization of Voltage level detector

Figure 2: Realizations of the Voltage level detector

A photo-detector converts incident light intensity into a voltage V_i . This voltage is applied to a voltage level detector, whose output V_o drives a switching circuit to turn a light ON or OFF, as shown in figure 1. The switching voltage is $V_{sw} = 4V$. [Hint: V_o is high/low for $V_i > V_{sw}$, and low/high for the opposite case.]

- The voltage level detector is first implemented using the circuit in figure 2a. Find the ratio R_1/R_2 , if $V_{ref} = -3V$, to meet the given specifications. State **briefly** what issues may occur if **noise** is introduced in V_i . [7]
- To address the issue in (a), the voltage level detector is redesigned using the circuit in 2b. If the **hysteresis width** is $160mV$, find R_2 & V_{ref} . Assume $R_1 = 2.2k\Omega$. [Hint: V_{sw} should be the midpoint of the thresholds] [7]
- Draw the **VTC** of the Schmitt Trigger with proper labels, showing the values of the **thresholds**, **bias voltages** & V_{sw} . [6]

Answer any TWO from the following

Question 2:

[15]

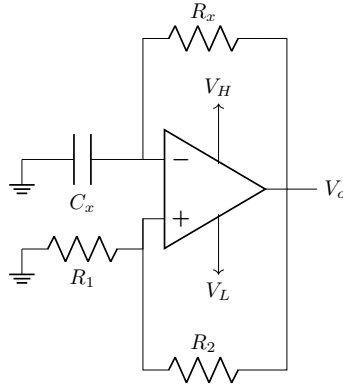


Figure 3: A Square Wave Generator

For the circuit in figure 3, $V_H = -V_L$. Time period of the generated square wave, $T = 4R_x C_x$.

(a) Find the ratio $R_1 : R_2$.

[8]

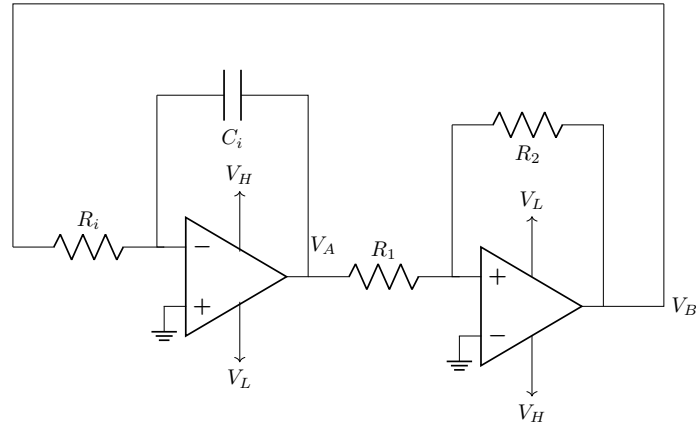


Figure 4: A Triangular Wave Generator

For the circuit in figure 4, $V_H = 13\text{ V}$, $V_L = -11\text{ V}$, $R_1 = 30\text{ k}\Omega$, $R_2 = 40\text{ k}\Omega$.

(b) Design the circuit to achieve an oscillation frequency of 1 kHz . [Hint: Assume a reasonable value for either R_i or C_i , and find the value of the other.] [7]

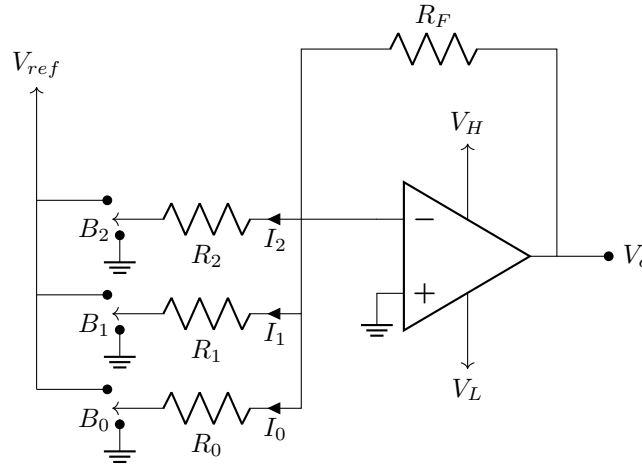


Figure 5: A 3-bit BWR DAC

For the BWR DAC shown in figure 5, $V_{ref} = -3V$, $R_0 = 2R_F$, $V_H = +12V$, $V_L = -12V$. Input bit sequence: $B_2B_1B_0$.

- Find the **step size** & the **MSB voltage** of the DAC. [3]
- Find V_o for the input cases such that, *at least two input branches conduct*. [5]
- Find I_1/I_2 & I_0/I_1 for the input bit sequence '101'. [4]
- Find the **maximum** possible value of V_{ref} , so that there are no duplicate outputs. [3]

Question 4:

[15]

Scenario-1: You are provided with a **3-bit ADC** circuit that will be used to convert an audio input signal to its digital equivalent. The audio input signal's amplitude ranges from $-7V$ to $+3V$. The ADC is a complete black box to you. Only two pieces of information are provided about the ADC.

- It is very fast
- One of the quantized levels is $-2V$

Scenario-2: Your friend uses a **3-bit Dual Slope ADC** circuit and converts the input's range to $0V$ to $10V$ by adding $+7V$ offset. The time constant τ of the integrator is $1.5ms$. Maximum allowable sampling frequency is $1.5kHz$.

- Context: Scenario 1:** What is the **type** of the ADC? Determine the **resolution**. [3]
- Context: Scenario 1:** Make a table listing the **quantization intervals** & corresponding **digital outputs**. What is the output associated with an input of $1.65V$? [5]
- Context: Scenario 2:** [7]
 - Find the integration phase duration (t_1) and the **clock frequency**. If the integrator's output is $-2.5V$, what is the corresponding **input voltage**?
 - To get the same digital output for an input of $1.65V$ as in (b), what should be the **reference voltage** of the Dual Slope ADC?